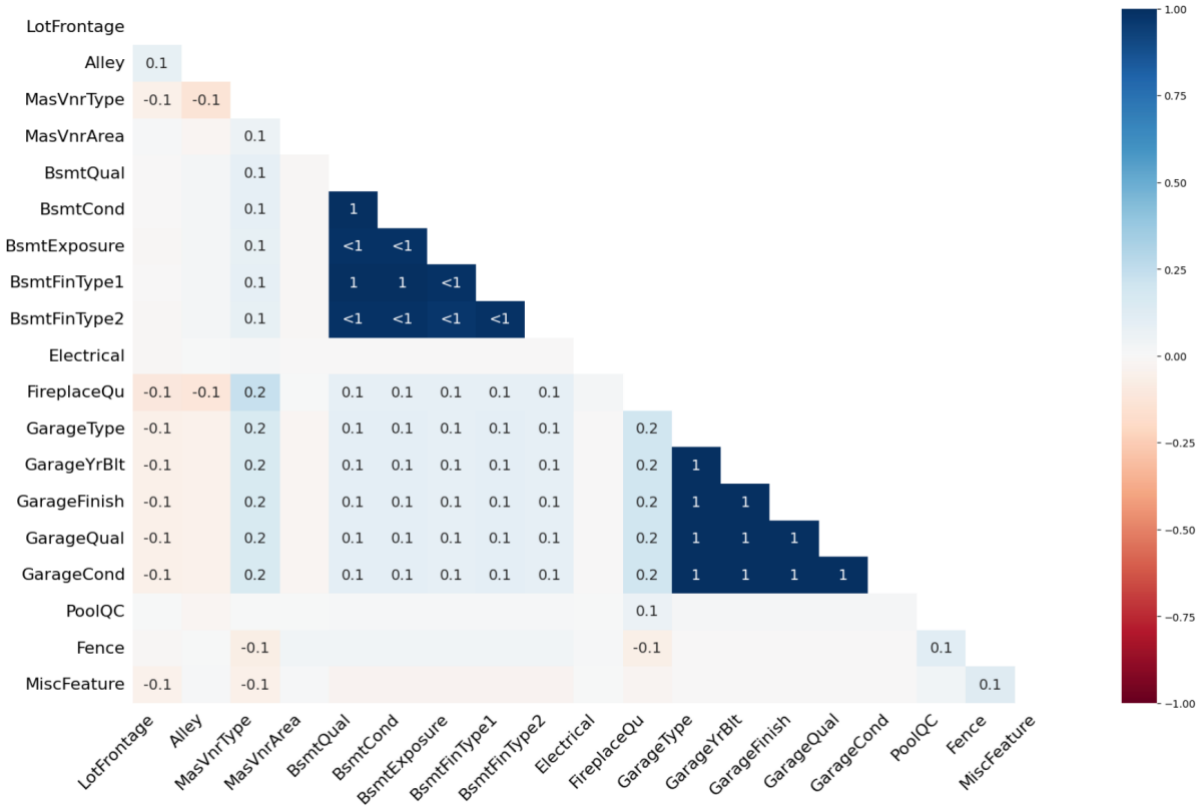
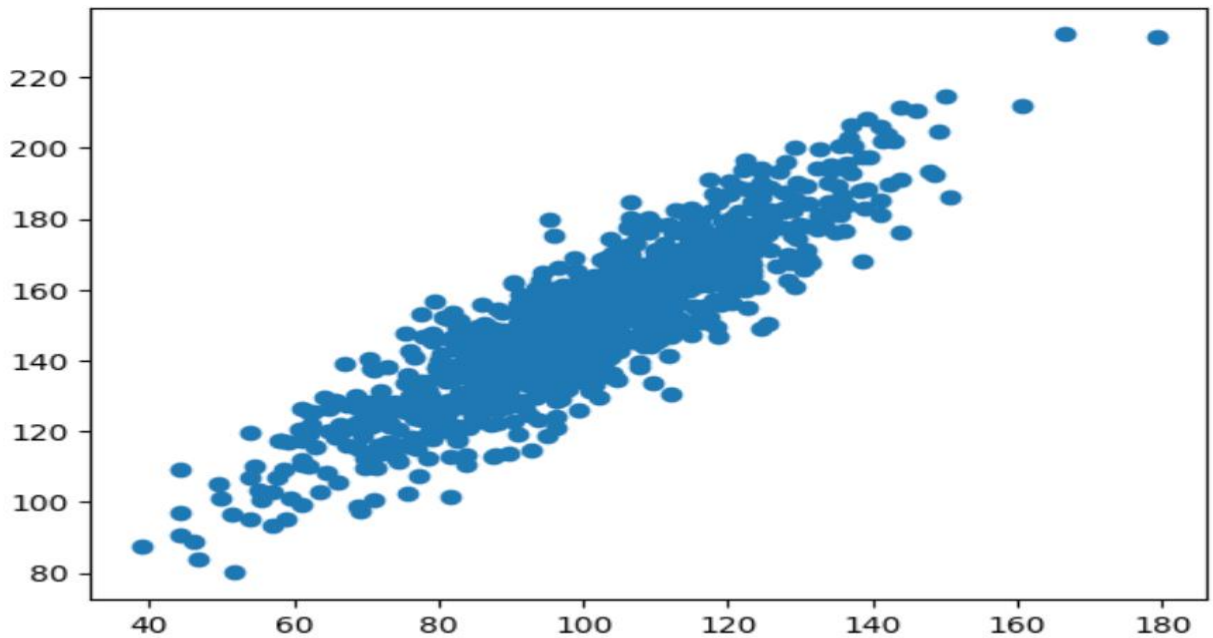
Artefact 1: ePortfolio Activities (Individual Learning Across Units)

1. Unit 2: A Tutorial on Exploratory Data Analysis: Heatmap



2. Unit 3: Covariance Pearson Correlation: Summary




```
data1: mean=100.776 stdv=19.620
data2: mean=151.050 stdv=22.358
Covariance: 389.755
Pearsons correlation: 0.888
```

3. Unit03 Linear Regression: Predicted Speed for 1337

```
from scipy import stats

x = [5,7,8,7,2,17,2,9,4,11,12,9,6]
y = [99,86,87,88,111,86,103,87,94,78,77,85,86]

slope, intercept, r, p, std_err = stats.linregress(x, y)

def myfunc(x):
  return intercept + slope * x

speed = myfunc(1337)

print(speed)
```

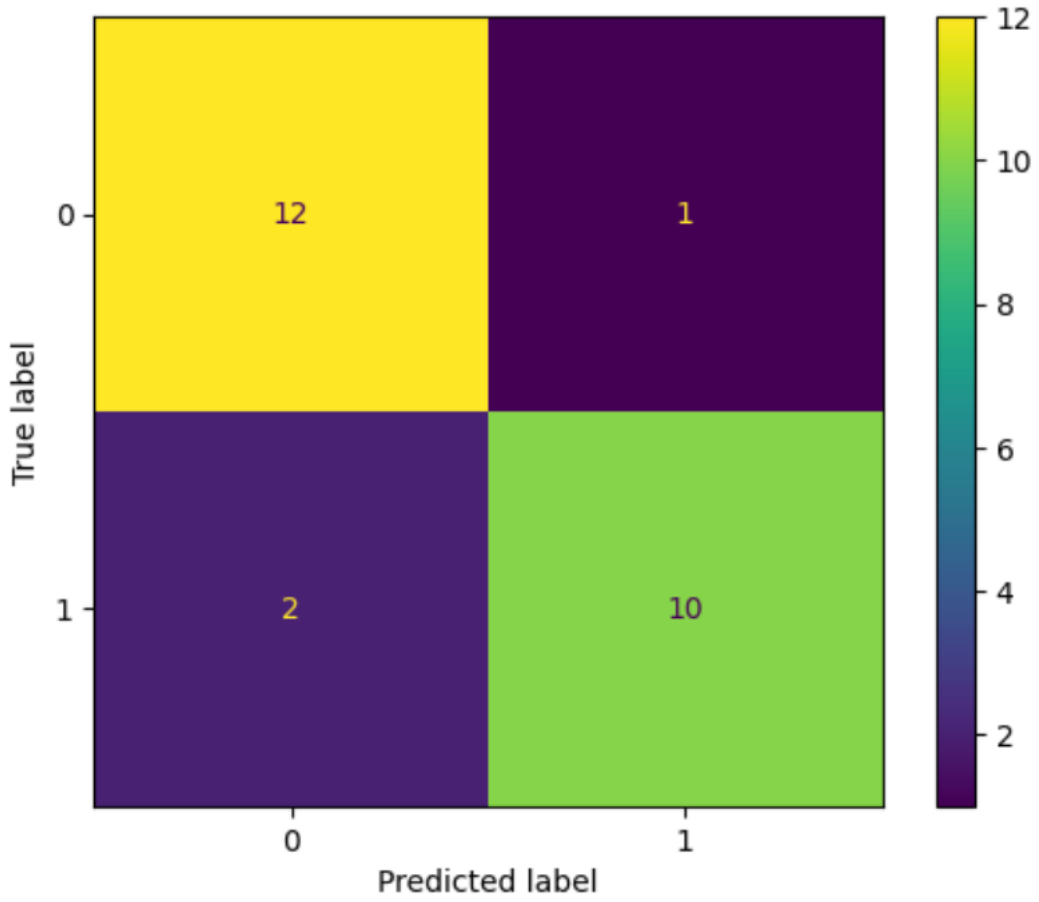
```
-2238.3657100809414
```

4. Unit03 Polynomial Regression: Prediction for speed at 1 AM

```
import numpy
from sklearn.metrics import r2_score

x = [1,2,3,5,6,7,8,9,10,12,13,14,15,16,18,19,21,22]
y = [100,90,80,60,60,55,60,65,70,70,75,76,78,79,90,99,99,100]

mymodel = numpy.poly1d(numpy.polyfit(x, y, 3))

speed = mymodel(1)
print(speed)
```

```
99.54274392967325
```

5. Unit11 Model Performance Measurement: Confusion Matrix



This artefact represents my individual learning progression across the module, captured through selected ePortfolio activities and Jupyter notebooks. It includes examples of data exploration, preprocessing, and early model implementation.

These activities demonstrate my progression from understanding basic dataset handling to applying and evaluating machine learning techniques. I developed practical skills in exploratory data analysis, feature selection, and implementing models such as regression and clustering, while also recognising their assumptions and limitations.

A key insight was the impact of dataset structure on model performance. Working with real data highlighted challenges such as missing values, weak correlations, and imbalance, reinforcing the need for careful preprocessing and critical interpretation.

Overall, this artefact demonstrates the development of my technical foundation and my shift towards a more critical and applied understanding of machine learning.